

Advanced (Habanero)

Q1. Simplify $2^3 \cdot 2^4$

- (A) 2^7
- (B) 2^6
- (C) 2^5
- (D) 2^4
- (E) 2^3

Q2. Simplify $(3^2)^4$

- (A) 3^8
- (B) 3^6
- (C) 3^5
- (D) 3^4
- (E) 3^3

Q3. Simplify $4^{\frac{6}{4^2}}$

- (A) 4^4
- (B) 4^5
- (C) 4^3
- (D) 4^2
- (E) 4^1

Q4. Simplify $x^3 \cdot x^2$

- (A) x^5
- (B) x^6
- (C) x^4
- (D) x^3
- (E) x^2

Q5. Simplify $(2x^2)^3$

- (A) $8x^6$
- (B) $8x^5$
- (C) $8x^4$
- (D) $4x^6$
- (E) $4x^5$

Q6. Simplify $\frac{x^8}{x^3}$

- (A) x^5
- (B) x^6
- (C) x^4
- (D) x^3
- (E) x^2

Q7. Simplify $(x^2)^4 \cdot x^3$

- (A) x^{11}
- (B) x^{10}
- (C) x^9
- (D) x^8
- (E) x^7

Q8. Simplify $\frac{(2x^2)^3}{(2x)^2}$

- (A) $2x^4$
- (B) $2x^3$
- (C) $2x^2$
- (D) x^4
- (E) x^3

Open Ended Questions (FRQ)

Q9. Simplify the expression $\frac{(3x^2)^2}{(3x)^2}$ and show your work.

Q10. Simplify the expression $(2x^3)^4 \cdot \frac{(x^2)^3}{x^4}$ and show your work.

Chef's Tips

- Read the questions carefully before answering.
- Show all your work for free response questions.
- Use proper mathematical notation.